

MAT 130

CHAPTER 7.3

INTEGRATING TRIGONOMETRIC FUNCTIONS

In Section 7.2, we derived reduction formulas for integrating positive integer powers of sine, cosine, tangent and secant. In this section, we will show how to work with those reduction formulas and we will discuss methods for integrating other kinds of integrals that involve trigonometric functions.

Integrating Powers of Sine and Cosine

Recall the following reduction formula

$$I_n = \int \sin^n x dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2} \text{ with } n \neq 0 \quad (1)$$

$$I_n = \int \cos^n x dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2} \text{ with } n \neq 0 \quad (2)$$

In the case, when $n = 2$, these formulas yield

$$\int \sin^2 x dx = -\frac{1}{2} \sin x \cos x + \frac{1}{2} x + c = \frac{1}{2} x - \frac{1}{4} \sin 2x + c \quad (3)$$

$$\int \cos^2 x dx = \frac{1}{2} \sin x \cos x + \frac{1}{2} x + c = \frac{1}{2} x + \frac{1}{4} \sin 2x + c \quad (4)$$

Alternative forms of these integration formulas can be derived from the trigonometric identities

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \quad \text{and} \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

In the case, when $n = 3$, formulas (1) and (2) yield

$$\int \sin^3 x dx = -\frac{1}{3} \sin^2 x \cos x + \frac{2}{3} \int \sin x dx = -\frac{1}{3} \sin^2 x \cos x - \frac{2}{3} \cos x + c \quad (5)$$

$$\int \cos^3 x dx = \frac{1}{3} \cos^2 x \sin x + \frac{2}{3} \int \cos x dx = \frac{1}{3} \cos^2 x \sin x + \frac{2}{3} \sin x + c \quad (6)$$

Formula (5) can be expressed in terms of *cosines* alone by using the identity $\sin^2 x = 1 - \cos^2 x$ and Formula (6) can be expressed in terms of *sines* alone by using the identity $\cos^2 x = 1 - \sin^2 x$.

Therefore, we get

$$\begin{aligned} \int \sin^3 x dx &= \int (\sin^2 x)(\sin x) dx = \int (1 - \cos^2 x)(\sin x) dx = \int \sin x dx - \int \cos^2 x \sin x dx \\ &= -\cos x + \frac{1}{3} \cos^3 x + c = \frac{1}{3} \cos^3 x - \cos x + c \\ \int \cos^3 x dx &= \int (\cos^2 x)(\cos x) dx = \int (1 - \sin^2 x)(\cos x) dx = \int \cos x dx - \int \sin^2 x \cos x dx \\ &= \sin x - \frac{1}{3} \sin^3 x + c \end{aligned}$$

Applying reduction formulas, we get

$$\int \sin^4 x dx = -\frac{1}{4} \sin^3 x \cos x + \frac{3}{4} \int \sin^2 x dx = -\frac{1}{4} \sin^3 x \cos x - \frac{3}{16} \sin 2x + \frac{3}{8} x + c$$

$$\int \cos^4 x dx = \frac{1}{4} \cos^3 x \sin x + \frac{3}{8} \int \cos^2 x dx = \frac{1}{4} \cos^3 x \sin x + \frac{3}{16} \sin 2x + \frac{3}{8} x + c$$

We can write $\sin 4x = 2 \sin 2x \cos 2x = 4 \sin x \cos x (2 \cos^2 x - 1) = 8 \cos^3 x \sin x - 4 \sin x \cos x$
 $= 4 \sin x \cos x (1 - 2 \sin^2 x) = 4 \sin x \cos x - 8 \sin^3 x \cos x$

Therefore, we get

$$\begin{aligned}\int \sin^4 x \, dx &= \frac{1}{32} (4 \sin x \cos x - 8 \sin^3 x \cos x) - \frac{1}{8} \sin x \cos x - \frac{3}{8} \sin x \cos x + \frac{3}{8} x + c \\ &= \frac{1}{32} \sin 4x - \frac{1}{2} \sin x \cos x + \frac{3}{8} x + c = \frac{3}{8} x - \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + c \\ \int \cos^4 x \, dx &= \frac{1}{32} (8 \cos^3 x \sin x - 4 \sin x \cos x) + \frac{1}{8} \sin x \cos x + \frac{3}{8} \sin x \cos x + \frac{3}{8} x + c \\ &= \frac{1}{32} \sin 4x + \frac{1}{2} \sin x \cos x + \frac{3}{8} x + c = \frac{3}{8} x + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + c\end{aligned}$$

Example 1

Find the volume V of the solid that is obtained when the region under the curve $y = \sin^2 x$ over the interval $[0, \pi]$ is revolved about the x -axis (Figure 7.3.1).

Solution: Using volume formula (method of disks, Chapter 6.2, page 425), we can write

$$\begin{aligned}V &= \int_0^\pi \pi \sin^4 x \, dx = \pi \left[\frac{3}{8} x - \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x \right]_0^\pi \\ &= \pi \left(\frac{3}{8} \pi - \frac{1}{4} \sin 2\pi + \frac{1}{32} \sin 4\pi \right) - 0 = \frac{3}{8} \pi^2.\end{aligned}$$

Ref. Theorem 5.6.1, Howard Anton (page 363)

Fundamental Theorem of Integral Calculus

If f is continuous on $[a, b]$ and if F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

See the Examples of Chapter 5.6

Definition (Chapter 5.2: The Indefinite Integral, page 322, Howard Anton)

A function F is called an antiderivative of a function f on a given open interval if $F'(x) = f(x)$ for all x in the interval.

For example, $f(x) = x^2$ has many antiderivatives such as

$$F_1(x) = \frac{1}{3} x^3, \quad F_2(x) = \frac{1}{3} x^3 + \ln 5, \quad F_3(x) = \frac{1}{3} x^3 + \sin 67, \quad F_4(x) = \frac{1}{3} x^3 + 2, \dots$$

Example 2 Evaluate (a) $\int \sin^4 x \cos^5 x \, dx$, (b) $\int \sin^4 x \cos^4 x \, dx$

Solution: (a) We can write $\int \sin^4 x \cos^5 x \, dx = \int \sin^4 x \cos^4 x \cos x \, dx$

Let $\sin x = u$. Then $\cos x \, dx = du$.

Therefore, $\int \sin^4 x \cos^5 x \, dx = \int u^4 (1 - u^2)^2 \, du = \int u^4 (1 - 2u^2 + u^4) \, du$

$$= \frac{1}{5} \sin^5 x - \frac{2}{7} \sin^7 x + \frac{1}{9} \sin^9 x + c$$

(b) We can write $\int \sin^4 x \cos^4 x dx = \int \left[\frac{1}{2} (1 - \cos 2x)^2 \right] \left[\frac{1}{2} (1 + \cos 2x)^2 \right] dx$

Let $2x = u$. Then $2 dx = du \Rightarrow dx = \frac{1}{2} du$.

Therefore, $\int \sin^4 x \cos^4 x dx = \frac{1}{16} \int (1 - \cos^2 2x)^2 dx = \frac{1}{16} \int \sin^4 2x dx = \frac{1}{32} \int \sin^4 u du$

$$= \frac{1}{32} \left(\frac{3}{8} u - \frac{1}{4} \sin 2u + \frac{1}{32} \sin 4u \right) + c = \frac{3}{128} x - \frac{1}{128} \sin 4x + \frac{1}{1024} \sin 8x + c$$

Example 3 Evaluate $\int \sin 7x \cos 3x dx$

Solution: We can write $\int \sin 7x \cos 3x dx = \frac{1}{2} \int (\sin 10x + \sin 4x) dx = -\frac{1}{20} \cos 10x - \frac{1}{8} \cos 4x + c$

Reduction formula for $\int \tan^n x dx$

$$\int \tan^n x dx = \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x dx$$

Reduction formula for $\int \sec^n x dx$

$$\int \sec^n x dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x dx$$

Hence, we get $\int \tan x dx = \ln |\sec x| + c$

$$\int \sec x dx = \ln |\sec x + \tan x| + c$$

$$\int \tan^2 x dx = \int (\sec^2 x - 1) dx = \tan x - x + c$$

$$\int \sec^2 x dx = \tan x + c$$

$$\int \tan^3 x dx = \frac{1}{2} \tan^2 x - \ln |\sec x| + c$$

$$\int \sec^3 x dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + c$$

Example 4 Evaluate (a) $\int \tan^2 x \sec^4 x dx$, (b) $\int \tan^3 x \sec^3 x dx$, (c) $\int \tan^2 x \sec x dx$

Solution: **(a)** We can write $\int \tan^2 x \sec^4 x dx = \int \tan^2 x (1 + \tan^2 x) \sec^2 x dx$

Let $\tan x = u$. Then $\sec^2 x dx = du$.

Therefore, $\int \tan^2 x \sec^4 x dx = \int u^2 (1 + u^2) du$

$$= \frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + c$$

(b) We can write $\int \tan^3 x \sec^3 x dx = \int \tan^2 x \sec^2 x (\sec x \tan x) dx$

Let $\sec x = u$. Then $\sec x \tan x dx = du$.

$$\begin{aligned}\text{Therefore, } \int \tan^3 x \sec^3 x \, dx &= \int (\sec^2 x - 1) \sec^2 x (\sec x \tan x) \, dx \\ &= \int (u^2 - 1)u^2 \, du = \frac{1}{5} \sec^5 x - \frac{1}{3} \sec^3 x + c\end{aligned}$$

$$\begin{aligned}\text{(c) We can write } \int \tan^2 x \sec x \, dx &= \int (\sec^2 x - 1) \sec x \, dx = \int \sec^3 x \, dx - \int \sec x \, dx \\ &= \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| - \ln |\sec x + \tan x| + c \\ &= \frac{1}{2} \sec x \tan x - \frac{1}{2} \ln |\sec x + \tan x| + c\end{aligned}$$

EXERCISE SET 7.3

2. Evaluate $\int \sin^5 3x \cos 3x \, dx$.

$$\begin{aligned}\text{Solution: We can write } \int \sin^5 3x \cos 3x \, dx &= \frac{1}{3} \int \sin^5 3x (3 \cos 3x) \, dx = \frac{1}{3} \times \frac{1}{6} \sin^6 3x + c \\ &= \frac{1}{18} \sin^6 3x + c\end{aligned}$$

4. Evaluate $\int \cos^3 at \, dt$.

$$\begin{aligned}\text{Solution: We can write } \int \cos^3 at \, dt &= \int (1 - \sin^2 at) \cos at \, dt = \int \cos at \, dt - \int \sin^2 at \cos at \, dt \\ &= \frac{1}{a} \sin at - \frac{1}{3a} \sin^3 at + c\end{aligned}$$

8. Evaluate $\int \sin^3 x \cos^3 x \, dx$.

$$\begin{aligned}\text{Solution: We can write } \int \sin^3 x \cos^3 x \, dx &= \int \sin^3 x (1 - \sin^2 x) \cos x \, dx \\ &= \int \sin^3 x \cos x \, dx - \int \sin^5 x \cos x \, dx \\ &= \frac{1}{4} \sin^4 x - \frac{1}{6} \sin^6 x + c\end{aligned}$$

9. Evaluate $\int \sin^2 t \cos^3 t \, dt$.

$$\begin{aligned}\text{Solution: We can write } \int \sin^2 t \cos^3 t \, dt &= \int \sin^2 t (1 - \sin^2 t) \cos t \, dt \\ &= \int \sin^2 t \cos t \, dt - \int \sin^4 t \cos t \, dt \\ &= \frac{1}{3} \sin^3 t - \frac{1}{5} \sin^5 t + c\end{aligned}$$

10. Evaluate $\int \sin^3 x \cos^2 x \, dx$.

$$\begin{aligned}\text{Solution: We can write } \int \sin^3 x \cos^2 x \, dx &= \int \sin^2 x \cos^2 x \sin x \, dx \\ &= \int (1 - \cos^2 x) \cos^2 x \sin x \, dx \\ &= \int \cos^2 x \sin x \, dx - \int \cos^4 x \sin x \, dx\end{aligned}$$

$$= -\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x + c$$

17. Evaluate $\int_0^{\pi/2} \cos^3 x \, dx$.

Solution: We can write $\int \cos^3 x \, dx = \sin x - \frac{1}{3} \sin^3 x$

$$\text{Hence } \int_0^{\pi/2} \cos^3 x \, dx = \left[\sin x - \frac{1}{3} \sin^3 x \right]_0^{\pi/2} = 1 - \frac{1}{3} = \frac{2}{3}$$

19. Evaluate $\int_0^{\pi/3} \sin^4 3x \cos^3 3x \, dx$.

Solution: We can write $\int \sin^4 3x \cos^3 3x \, dx = \int \sin^4 3x (1 - \sin^2 3x) \cos 3x \, dx$
 $= \int \sin^4 3x \cos 3x \, dx - \int \sin^6 3x \cos 3x \, dx$

$$\text{Hence } \int_0^{\pi/3} \sin^4 3x \cos^3 3x \, dx = \left[\frac{1}{15} \sin^5 3x - \frac{1}{21} \sin^7 3x \right]_0^{\pi/3} = 0 \text{ since } \sin \pi = \sin 0 = 0.$$

25. Evaluate $\int e^{-x} \tan(e^{-x}) \, dx$.

Solution: We can write $\int e^{-x} \tan(e^{-x}) \, dx = -\int \tan u \, du$ since for $e^{-x} = u \Rightarrow -e^{-x} dx = du$

Therefore, we get $\int e^{-x} \tan(e^{-x}) \, dx = \ln |\cos(e^{-x})| + c$

30. Evaluate $\int \tan^5 x \sec^4 x \, dx$.

Solution: We can write $\int \tan^5 x \sec^4 x \, dx = \int \tan^5 x (1 + \tan^2 x) \sec^2 x \, dx$

Let $\tan x = u$. Then $\sec^2 x \, dx = du$.

$$\text{Therefore, } \int \tan^5 x \sec^4 x \, dx = \int u^5 (1 + u^2) \, du = \frac{1}{6} \tan^6 x + \frac{1}{8} \tan^8 x + c$$

33. Evaluate $\int \sec^5 x \tan^3 x \, dx$.

Solution: We can write $\int \sec^5 x \tan^3 x \, dx = \int \sec^4 x (\sec x \tan x) (\sec^2 x - 1) \, dx$

Let $\sec x = u$. Then $\sec x \tan x \, dx = du$.

$$\text{Therefore, } \int \sec^5 x \tan^3 x \, dx = \int u^4 (u^2 - 1) \, du = \frac{1}{7} \sec^7 x - \frac{1}{5} \sec^5 x + c$$

35. Evaluate $\int \tan^4 x \sec x \, dx$.

Solution: We can write $\int \tan^4 x \sec x \, dx = \int (\sec^2 x - 1)^2 \sec x \, dx$
 $= \int (\sec^4 x - 2\sec^2 x + 1) \sec x \, dx = \int (\sec^5 x - 2\sec^3 x + \sec x) \, dx$
 $= \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} \int \sec^3 x \, dx - 2 \int \sec^3 x \, dx + \ln |\sec x + \tan x|$

$$\begin{aligned}
&= \frac{1}{4} \sec^3 x \tan x - \frac{5}{4} \left[\frac{1}{2} \sec x \tan x + \frac{1}{2} \int \sec x \, dx \right] + \ln |\sec x + \tan x| \\
&= \frac{1}{4} \sec^3 x \tan x - \frac{5}{8} \sec x \tan x - \frac{5}{8} \ln |\sec x + \tan x| + \ln |\sec x + \tan x| + c \\
&= \frac{1}{4} \sec^3 x \tan x - \frac{5}{8} \sec x \tan x + \frac{3}{8} \ln |\sec x + \tan x| + c
\end{aligned}$$

39. Evaluate $\int \sec^4 x \, dx$.

Solution: We can write $\int \sec^4 x \, dx = \int (1 + \tan^2 x) \sec^2 x \, dx$

$$= \int \sec^2 x \, dx + \int \tan^2 x \sec^2 x \, dx = \tan x + \frac{1}{3} \tan^3 x + c$$

Reduction formula for $\int \sec^n x \, dx$

$$\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

40. Evaluate $\int \sec^5 x \, dx$.

Solution: Using reduction formula, we get

$$\begin{aligned}
\int \sec^5 x \, dx &= \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} \int \sec^3 x \, dx \\
&= \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} \left[\frac{1}{2} \sec x \tan x + \frac{1}{2} \int \sec x \, dx \right] \\
&= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \ln |\sec x + \tan x| + c
\end{aligned}$$

Reduction formula for $\int \tan^n x \, dx$

$$\int \tan^n x \, dx = \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x \, dx$$

42. Evaluate $\int \tan^4 x \, dx$.

Solution: Using reduction formula, we get

$$\begin{aligned}
\int \tan^4 x \, dx &= \frac{1}{3} \tan^3 x - \int \tan^2 x \, dx = \frac{1}{3} \tan^3 x - \int (\sec^2 x - 1) \, dx \\
&= \frac{1}{3} \tan^3 x - \tan x + x + c
\end{aligned}$$

47. Evaluate $\int_0^{\pi/2} \tan^5(x/2) \, dx$.

Solution: Let $x/2 = u$. Then $dx = 2du$.

Also, when $x = 0$ then $u = 0$ and when $x = \pi/2$ then $u = \pi/4$.

Therefore, we get $\int_0^{\pi/4} \tan^5(x/2) dx = 2 \int_0^{\pi/4} \tan^5 u du$

Using reduction formula, we get

$$\begin{aligned} \int \tan^5 u du &= \frac{1}{4} \tan^4 u - \int \tan^3 u du = \frac{1}{4} \tan^4 u - \left[\frac{1}{2} \tan^2 u - \int \tan u du \right] \\ &= \frac{1}{4} \tan^4 u - \frac{1}{2} \tan^2 u - \ln |\cos u| \end{aligned}$$

$$\begin{aligned} \text{Hence, } \int_0^{\pi/4} \tan^5(x/2) dx &= \frac{1}{2} \left[\tan^4 u \right]_0^{\pi/4} - \left[\tan^2 u \right]_0^{\pi/4} - 2 \left[\ln |\cos u| \right]_0^{\pi/4} = \frac{1}{2} (1-0) - (1-0) - 2 \ln \left(\frac{1}{\sqrt{2}} \right) \\ &= -\frac{1}{2} + \ln 2 \end{aligned}$$

60. Find the volume of the solid generated when the region enclosed by $y = \tan x$, $y = 1$ and $x = 0$ is revolved about the x -axis.

Solution: Given $y = \tan x$, $y = 1$

Now $y = \tan x \Rightarrow x = \pi/4$

$$\text{Hence, volume, } V = \int_0^{\pi/4} \pi(1 - \tan^2 x) dx = \int_0^{\pi/4} \pi(2 - \sec^2 x) dx = \pi [2x - \tan x]_0^{\pi/4} = \pi \left(\frac{\pi}{2} - 1 \right).$$

61. Find the volume of the solid that results when the region enclosed by $y = \cos x$, $y = \sin x$, $x = 0$ and $x = \frac{\pi}{4}$ is revolved about the x -axis.

Solution: Given $y = \cos x$, $y = \sin x$, $x = 0$ and $x = \frac{\pi}{4}$.

$$\text{Volume of the generated solid, } V = \int_0^{\pi/4} \pi(\cos^2 x - \sin^2 x) dx = \pi \int_0^{\pi/4} \cos 2x dx = \frac{\pi}{2} [\sin 2x]_0^{\pi/4} = \frac{\pi}{2}.$$

62. The region bounded below by the x -axis and above by the portion of $y = \sin x$ from $x = 0$ to $x = \pi$ is revolved about the x -axis. Find the volume of the resulting solid.

Solution: Given $y = \sin x$ from $x = 0$ to $x = \pi$.

$$\text{Volume of the resulting solid, } V = \int_0^{\pi} \pi(\sin^2 x) dx = \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi} = \frac{\pi}{2} (\pi - 0) = \frac{\pi^2}{2}.$$